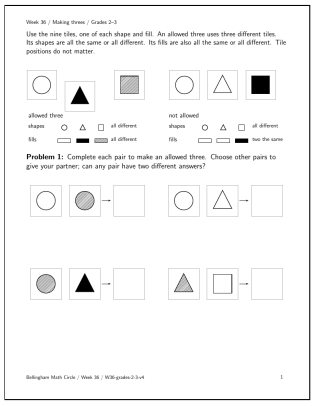
Week 36 / Making threes / Problem menu

for adults: pick, then print


Week 36 / Making threes / Grades 2–3

Use the nine tiles, one of each shape and fill. An allowed three uses three different tiles. Its shapes are all the same or all different. Its fills are also all the same or all different. Tile positions do not matter.

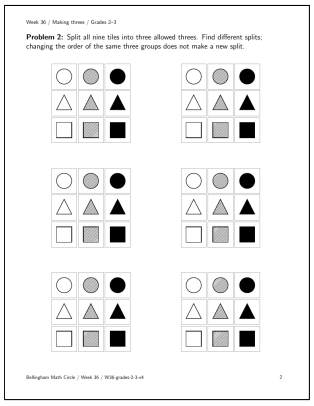
allowed three
shapes all different
fills all different

not allowed
shapes all different
fills two the same

Problem 1: Complete each pair to make an allowed three. Choose other pairs to give your partner; can any pair have two different answers?

Bellingham Math Circle / Week 36 / W36-grades-2-3-v4 1

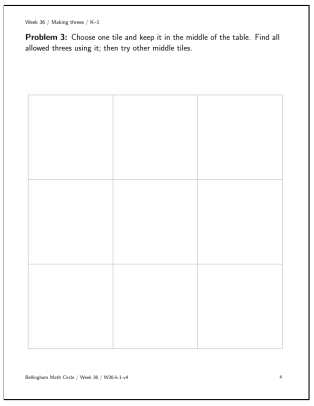
Complete each pair
K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1
first route
wording differs by band


Week 36 / Making threes / Grades 2–3

Problem 2: Split all nine tiles into three allowed threes. Find different splits; changing the order of the same three groups does not make a new split.

Bellingham Math Circle / Week 36 / W36-grades-2-3-v4 2

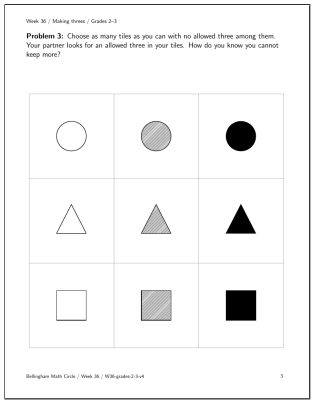
Split the nine tiles into three allowed threes
K–1 P2 · Grades 2–3 P2
first route
wording differs by band


Week 36 / Making threes / K–1

Problem 3: Choose one tile and keep it in the middle of the table. Find all allowed threes using it; then try other middle tiles.

Bellingham Math Circle / Week 36 / W36-k-1-v4 4

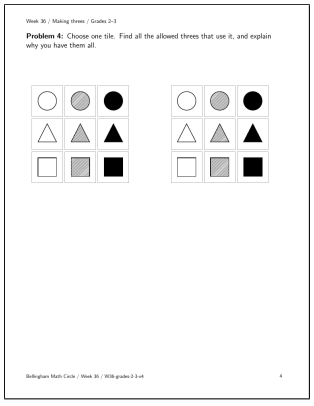
All the allowed threes through a middle tile (K-1)
K–1 P3
return route


Week 36 / Making threes / Grades 2–3

Problem 3: Choose as many tiles as you can with no allowed three among them. Your partner looks for an allowed three in your tiles. How do you know you cannot keep more?

Bellingham Math Circle / Week 36 / W36-grades-2-3-v4 3

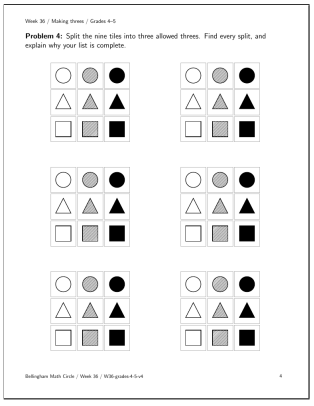
The most tiles with no allowed three
K–1 P4 · Grades 2–3 P3 · Grades 4–5 P2
first route
wording differs by band


Week 36 / Making threes / Grades 2–3

Problem 4: Choose one tile. Find all the allowed threes that use it, and explain why you have them all.

Bellingham Math Circle / Week 36 / W36-grades-2-3-v4 4

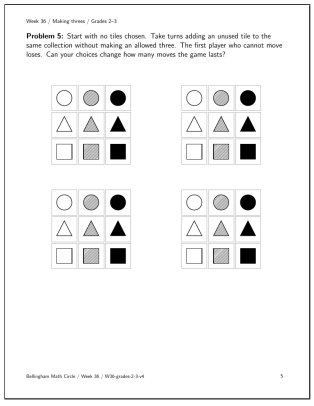
All the allowed threes through one tile
Grades 2–3 P4 · Grades 4–5 P3
return route
wording differs by band


Week 36 / Making threes / Grades 4–5

Problem 4: Split the nine tiles into three allowed threes. Find every split, and explain why your list is complete.

Bellingham Math Circle / Week 36 / W36-grades-4-5-v4 4

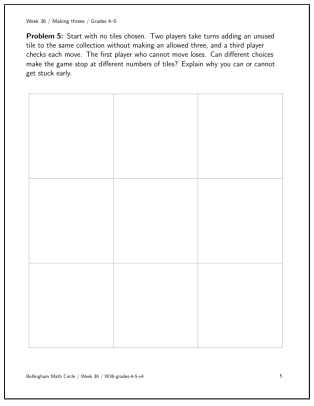
Every split, and why the list is complete
Grades 4–5 P4
return route


Week 36 / Making threes / Grades 2–3

Problem 5: Start with no tiles chosen. Take turns adding an unused tile to the same collection without making an allowed three. The first player who cannot move loses. Can your choices change how many moves the game lasts?

Bellingham Math Circle / Week 36 / W36-grades-2-3-v4 5

The adding game: can choices change its length?
Grades 2–3 P5
return route


Week 36 / Making threes / Grades 4–5

Problem 5: Start with no tiles chosen. Two players take turns adding an unused tile to the same collection without making an allowed three, and a third player checks each move. The first player who cannot move loses. Can different choices make the game stop at different numbers of tiles? Explain why you can or cannot get stuck early.

Bellingham Math Circle / Week 36 / W36-grades-4-5-v4 5

Can you get stuck early?
Grades 4–5 P5
return route